

NEO Mathematics Learner Guide

Pythagoras and the Hidden Length

How can a right-angled triangle reveal a length we cannot measure directly?

What are we exploring?

This unit is about finding hidden lengths. You will notice what is special about right-angled triangles, explore squares on the sides, build the Pythagorean relationship, decide when the theorem is allowed and use it to model indirect measurement.

Your Stepping Stones

Notice right-angled triangles and the hypotenuse.

Connect the areas of squares on the three sides.

Translate diagrams into $a^2 + b^2 = c^2$.

Calculate a hypotenuse or a shorter side.

Choose when Pythagoras is valid.

Check whether an answer makes sense.

Model a hidden measurement in a real or invented context.

A possible way through

Use the Unit Route page to open the next useful session. Sessions called **Reconnection** are optional routes. You may use one if squares, roots or angle facts are blocking the current idea. The aim is to reconnect and return.

You do not need to complete every session in one sitting or prove everything at once.

When you feel stuck

1. Find the right angle.
2. Find the side opposite it. That is the hypotenuse.
3. Ask whether you are finding the whole square-area total or one missing part.
4. Square first. Add or subtract the areas. Square root last.
5. Check whether the length is a sensible size.

Using support is part of learning. It is not cheating.

The 15 session opportunities

- 01. The hidden length** - What can a right-angled triangle reveal that we cannot measure directly?
- 02. Squares on the sides** - Why might the areas of three squares tell us about one triangle?
- 03. Reconnection: squares, roots and number structure** - Which number facts make hidden lengths easier to see?
- 04. From picture to equation** - How can a diagram become an equation without losing its meaning?
- 05. Finding the longest side** - How can two known lengths unlock the hypotenuse?
- 06. When the unknown is not the longest side** - What changes when the missing side is one of the shorter sides?
- 07. Does the answer make sense?** - Can an impossible answer look mathematically tidy?
- 08. Reconnection: angles and right triangles** - Which angle facts unlock the diagram?
- 09. Is Pythagoras allowed here?** - What tempting diagram features can make us use the wrong theorem?
- 10. Measuring what is hard to reach** - How can a hidden length support a real decision?
- 11. One diagonal, two routes** - Which route through the problem is clearest or most efficient?
- 12. A diagonal inside a shape** - Can one hidden right triangle be found inside a larger shape?
- 13. From two dimensions towards three** - Can a 3D distance be found using two linked right triangles?
- 14. What can I now see that was hidden before?** - Which connection in this unit is stronger—and which one needs another route?
- 15. How far can indirect measurement take me?** - How can I design and defend a model for measuring something indirectly?

The Scratchpad

The Scratchpad stays inside each lesson. You can use pen colours, Sticky Notes, undo/redo and a square grid. You may also use paper or another way of recording your thinking.

Useful colour idea

Try one colour for the right angle, another for the hypotenuse, and a third for the calculation route. This is optional.

How will I know I am making progress?

"I can find the hypotenuse."

"I can explain why the square areas matter."

"I can decide whether Pythagoras is allowed."

"I can find a hidden length and check whether it makes sense."

"I can build a right-triangle model for a measurement question."

You do not need all of these to be true yet. They are signs of where the learning is heading.

Before you finish a session

One thing I noticed...

One thing that feels clearer...

One thing I may reconnect...

The support that helped me most was...

Final portfolio piece

Your final challenge is to design an Indirect Measurement Portfolio Piece. Choose a hidden length, build a right-triangle model, calculate, check the answer, and explain one assumption or limitation.